DSAI 490 – Assignment 1
Representation Learning with Autoencoders (AE & β-VAE) on Medical MNIST

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1 Introduction and dataset

This work implements and compares a convolutional Autoencoder (AE) and a convolutional β-Variational Autoencoder (β-VAE) on the Medical MNIST dataset, which contains six classes of 64×64 grayscale medical images (AbdomenCT, BreastMRI, CXR, ChestCT, Hand, HeadCT). Data is split 80/10/10 train/validation/test and streamed through a `tf.data` pipeline (JPEG decoding, resize, $[0, 1]$ scaling, shuffle, batch, prefetch). Both models train for 30 epochs with Adam (10^{-3}) and batch size 128 on a single Kaggle GPU. The full codebase, trained weights, and all figures are reproduced by a single Jupyter notebook.

2 Model architectures

Autoencoder. A convolutional encoder with three `Conv2D + MaxPooling2D` blocks (filters 32→16→16) compresses 64×64×1 inputs to an 8×8×16 feature map (1024 latent dimensions). The decoder is a symmetric stack of three `Conv2DTranspose` layers with stride 2, followed by a final 3×3 sigmoid `Conv2D`. Loss is pixelwise binary cross-entropy.

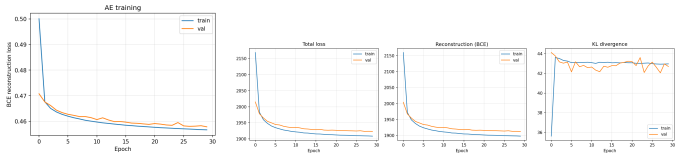
β-Variational Autoencoder. The encoder uses three strided `Conv2D` layers (32→64→64), a 128-unit dense layer, and two parallel dense heads producing $\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2 \in \mathbb{R}^{16}$. The reparameterisation trick $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, yields a differentiable sample that the decoder (dense → reshape → three `Conv2DTranspose` → sigmoid) reconstructs. The objective is the evidence lower bound with a scalar Kullback–Leibler weight:

$$\mathcal{L} = \underbrace{\mathbb{E}_q[\mathrm{BCE}(x, \hat{x})]}_{\text{reconstruction}} + \beta \cdot \underbrace{\mathrm{KL}(q(z \mid x) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I}))}_{\text{regulariser}},$$

with KL computed in closed form. We set $\beta = 0.25$: a pilot run at $\beta = 1$ exhibited partial posterior collapse (KL $\approx$ 27 nats) and severely blurred reconstructions, a known failure mode on datasets where BCE summed over pixels dominates the KL term. Down-weighting KL recovers encoding capacity; lowering β *raises* KL (observed $\approx$ 43 nats) because the encoder is allowed to move further from the prior.

Key difference. The AE has a deterministic 1024-dim code with no prior regularisation and is optimised only for reconstruction. The β-VAE has a probabilistic 16-dim code actively pulled toward a unit-Gaussian prior. The VAE therefore trades code capacity and class-wise separability for a continuous, sampleable latent — the price of a real generative model.

3 Results



(a) AE (BCE) (b) VAE: total / reconstruction (BCE) / KL

Figure 1: Training dynamics. AE BCE drops from 0.50 to 0.46 and plateaus after ∼20 epochs. The VAE reconstruction term drops from ∼2160 to ∼1900, while KL rises from its initialisation and stabilises at ∼43 nats — the encoder learning to carry information rather than collapsing to the prior.

Reconstruction (Fig. 2). Table 1 shows the AE achieves lower test pixel error than the β-VAE, as expected: the AE has 1024 unregularised dimensions versus 16 regularised ones and is optimised for exactly this objective. Both models reproduce coarse anatomy faithfully (hand fingers, chest cavity, skull silhouette); fine texture — trabecular bone in the Hand, parenchymal detail in HeadCT, peritoneal structure in AbdomenCT — is partially lost. The VAE output is visibly smoother, the canonical "VAE blur" driven by the 16-dim KL-regularised code and the Gaussian output likelihood.

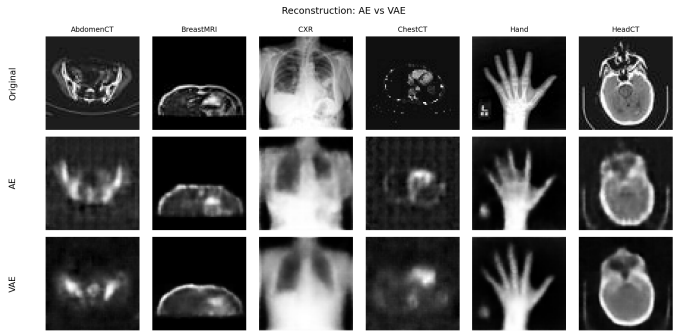


Figure 2: Reconstruction, one test example per class. Top: original. Middle: AE. Bottom: β-VAE. Both preserve global structure; both soften high-frequency texture.

	Test MSE	Test MAE
AE	0.00429	0.0355
β-VAE	0.00699	0.0449

Table 1: Pixel-wise reconstruction error on the held-out test set.

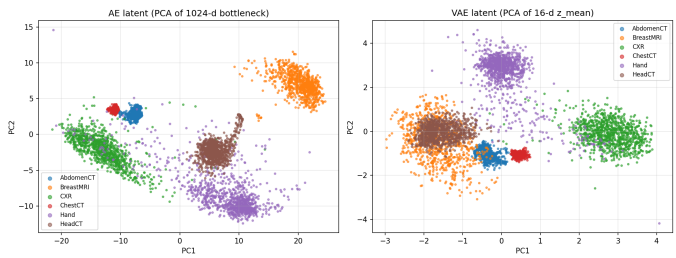


Figure 3: 2D PCA of test-set latents. AE (left): 1024-dim deterministic bottleneck yields tight, well-separated class clusters. β-VAE (right): 16-dim $\boldsymbol{\mu}$-space is compact and centered near the origin (by prior regularisation) with visible class structure but more inter-class overlap.

Latent space (Fig. 3). The AE bottleneck places BreastMRI, Hand, and the CT group in clearly distinct regions; this is precisely the structure a downstream classifier wants. The VAE $\boldsymbol{\mu}$-space is more compressed and more overlapping — an unavoidable consequence of KL

regularisation pulling every posterior toward the same prior. The VAE latent is nonetheless densely and continuously populated near the origin, which is what makes the *prior* distribution a meaningful generative target.

Figure 4: 64 samples $\hat{x} = \mathrm{dec}(z),\ z \sim \mathcal{N}(0, I)$. Several samples are class-identifiable (hands in rows 2 and 4, chest-cavity silhouettes in rows 1 and 5); others are class-ambiguous blends typical of a small convolutional VAE with a unit-Gaussian prior.

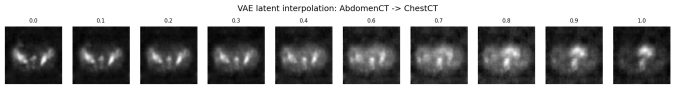


Figure 5: Linear interpolation in $\boldsymbol{\mu}$-space between an AbdomenCT and a ChestCT encoding. Smooth pixel-level transitions (no discrete jumps) indicate a continuous latent manifold, even though the intermediate frames remain visually ambiguous on this dataset.

Sample generation and interpolation. Figure 4 shows that the β-VAE has learned a generative distribution: hands and chest silhouettes emerge recurrently across the grid, confirming class structure in the prior. Samples are blurry — a property bounded by the Gaussian likelihood and small latent, not by training length. Richer priors (VQ-VAE, diffusion) are required for sharp medical-image synthesis at substantially higher cost. Figure 5 confirms that the latent manifold is smooth: no discontinuities appear along the path, a prerequisite for using the latent for semantic editing.

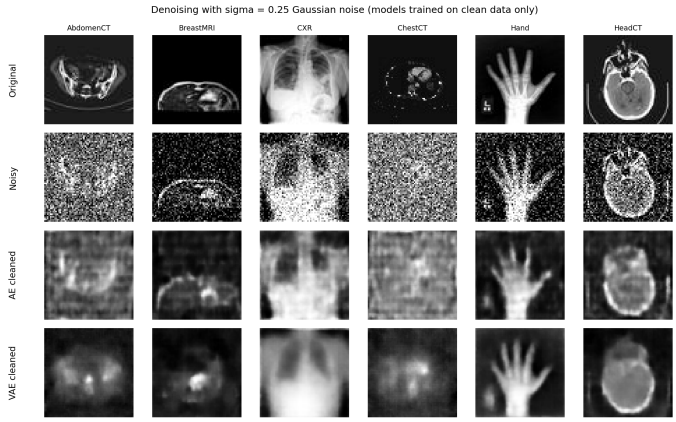


Figure 6: Both models, trained on clean data only, are fed inputs corrupted with σ=0.25 Gaussian noise. Both suppress most of the noise — an *emergent* property of the low-dimensional bottleneck. The β-VAE output is notably cleaner than the AE output, a direct consequence of the prior-regularised latent acting as a stronger implicit smoother.

Denoising (Fig. 6). Although neither model was trained on noisy inputs, both partly denoise σ=0.25 Gaussian corruption because the bottleneck cannot efficiently represent noise. The VAE visibly outperforms the AE here — an interesting inversion of the reconstruction ranking: the KL-regularised latent acts as a stronger implicit prior against perturbations, whereas the AE's higher-capacity code preserves more of the corruption. A purpose-trained denoising AE with noisy-input / clean-target pairs would outperform both but was out of scope.

4 Conclusions

The AE and β-VAE optimise different objectives and produce different latent geometries. On Medical MNIST, the AE wins on pixel-accurate reconstruction (39% lower MSE) and on class separability in latent space, making it the better *feature extractor*. The β-VAE wins on sample generation, latent interpolation, and (unexpectedly) denoising robustness, making it the better *generative model*. Neither model is universally superior; the architectural and loss-function choices encode which downstream capability is prioritised. The β=0.25 weighting was essential to avoid partial posterior collapse; this is a practical tuning point that the canonical β=1 literature tends to understate on small grayscale datasets.